

FINAL PROJECT SUMMARY: Is the Juice Worth the Squeeze?

An Educated Guess on the Payoff of Academia

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December 5, 2025

Project Aim: The present analysis investigates how institutional characteristics of U.S. colleges and universities relate to graduates' economic outcomes. We explored which school-level features best predict graduates' median earnings ten years after enrollment. Given the rising cost of higher education, increasing student loan debt, and ongoing debate surrounding the economic value of a college degree, this analysis seeks to clarify which institutional factors are associated with stronger long-term financial outcomes. Ultimately, these insights may help prospective students make more informed decisions about college choice and prompt institutions and policymakers to consider which campus characteristics most meaningfully support students' economic mobility.

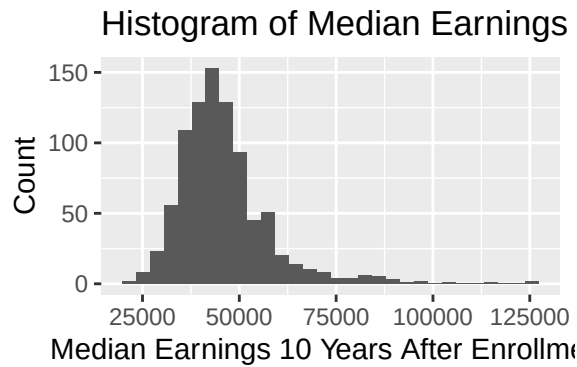
Variables: The response variable is median income ten years after enrollment (MD_EARN_WNE_P10). The explanatory variables we will be using to predict median earnings 10 years after enrollment are admissions rate (ADM_RATE), median verbal and quantitative SAT scores (SATVRMID; SATMTMID), total undergraduate enrollment (UGDS), average cost of attendance (COSTT4_A), student-faculty ratio (STUFACR) and median family income (MD_FAMINC).

Initial Hypotheses: Our initial hypotheses were that median SAT scores, median family incomes, and the cost of tuition would be positively correlated with median earnings ten years after entering the institution. We also hypothesized that school acceptance rates, and student to faculty ratio would be negatively associated with median earnings ten years after entering the institution. In our preliminary analyses, we found that these all ended up being true, at varying strengths of correlation coefficients.

A note on external validity: Although the College Scorecard dataset includes nearly 6,000 institutions, a substantial proportion do not report the key variables of interest to us, such as SAT scores, admissions rate, and student family income. As a result, our analytic sample includes only the subset of schools with complete information for our chosen variables (Total Complete Data N = 1173, Training N = 879). It is likely that this missing data is not random; schools that do not report comprehensive data may be more likely to have open enrollment, or have limited administrative capacity. Therefore, our conclusions apply primarily to institutions that report comprehensive admissions and outcome metrics.

Univariate EDA:

Response Variable Summary: Median Earnings 10 Years After Enrollment is unimodal and roughly normally distributed, with a rightward skew. The median is 43,600 and the mean is 45,821.73. The middle 50% of colleges have earnings between 38400 and 50100 (IQR approximately 13,000), with a large standard deviation of 12076.79, indicating substantial variations in Median 10 Year Earnings between colleges. There are several potential outliers on the right tail, which will be accounted for in the final model.

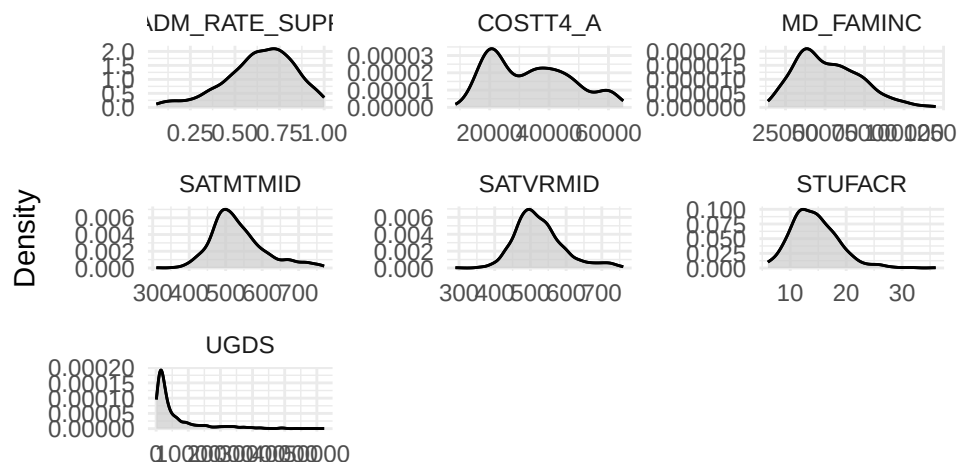


##	min	Q1	median	Q3	max	mean	sd	n	missing
##	20600	38400	43600	50100	124700	45821.73	12076.79	879	0

Response Variable Summary: Median Earnings 10 Years After Enrollment is unimodal and roughly normally distributed, with a rightward skew. The median is 43,600 and the mean is 45,821.73. The middle 50% of colleges have earnings between 38,400 and 50,100 (IQR approximately 13,000), with a large standard deviation of 12,076.79, indicating substantial variations in Median 10 Year Earnings between colleges. There are several potential outliers on the right tail, which will be accounted for in the final model.

Predictor Variables Summary: To characterize the distributions of our predictor variables, we examined histograms, density plots, and key summary statistics (median, range, and IQR). **Right-Skewed Variables** Several variables display a clear right skew, indicating a concentration of observations on the lower end and a long upper tail. This group includes Average cost of attendance, Median family income, Student-faculty ratio, Undergraduate enrollment (certificate-seeking students). All predictor variables except average cost of attendance appear unimodal. Each shows potential high-end outliers, and formal outlier detection is necessary. Average Cost of Attendance: Bimodal; median = 32,707; min = 8,461; max = 64,988; range = 56,527; IQR = 21,993.5. Median Family Income: Median = 49,259; min = 13,833; max = 120,000; range = 70,741; IQR = 29,391.5. Student-Faculty Ratio: Median = 14; range = 6 to 36 (range = 30); IQR = 6. **Left-Skewed Variable** Only admissions rate shows left skew, with most schools admitting moderate to high percentages of applicants and fewer highly selective institutions. It is unimodal with no strong signs of outliers Admission Rate: Median = 0.66; Min = 0.0596 (approx 6%), max = 1.00 (100%); Range = 0.94; IQR = 0.25 **Approximately Normal Variables** Two variables appear roughly symmetric, unimodal and bell-shaped, with no visible outliers: SAT Math: Mean = 532.95; range = 310–770; IQR = 83.5 SAT Verbal Mean = 524.66; range = 290–760; IQR = 79.5

Density Plots for Predictor Variables (Training Set)

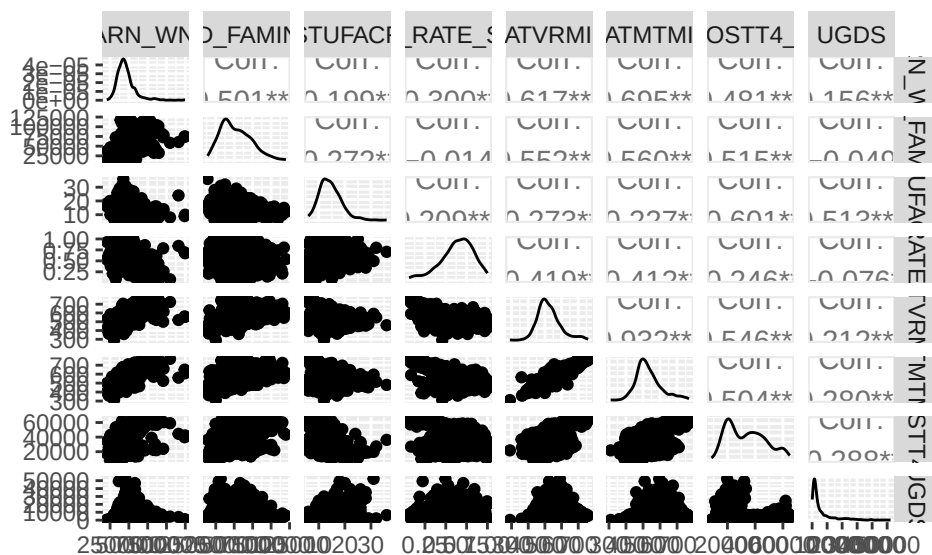


Bivariate EDA: The graph above allowed us to examine the bivariate relationship between each explanatory variable and median earnings ten years after enrollment (MD_EARN_WNE_P10). We first produced a

GGpairs plot to view scatterplots, density plots, and correlation coefficients among each of the 6 explanatory variables (ADM_RATE, SATVRMID, SATMTMID, UGDS, COSTT4_A, MD_FAMINC) and the response variable (MD_EARN_WNE_P10.) This provides an initial assessment of which institutional characteristics appear most strongly associated with post-graduation earnings, and it allows us to visually inspect distribution patterns and potential outliers. We also created graphs for each individual predictor and MD_EARN_WNE_P10, though we omitted them for the sake of space.

Median Earnings 10 Years After Enrollment was positively correlated with Math and Verbal SAT scores ($r = 0.695$, $r = 0.617$, respectively). It was also positively correlated with median family income ($r = 0.501$), though we can see homoscedasticity. This may be partially due to the fact that most top academic schools have extensive scholarship opportunities for students of a range of incomes, and end up producing high earners. Additionally, MD_EARN_WNE_P10 was positively correlated with cost of attendance ($r = 0.481$). It was weakly negatively correlated with student-faculty ratio ($r = -0.199$) and admissions rate ($r = -0.300$). Though notably, for highly selective schools (admissions rate $< 25\%$), there appears to be a moderate negative relationship where lower acceptance rates are correlated with higher future earnings. However, once the admissions rate exceeds roughly 25%, the slope becomes almost flat.

The GGpairs matrix also helped us to identify multicollinearity among predictors, such as SAT verbal and SAT math scores ($r = 0.932$).



Our Current Model: First, we ran a “kitchen sink” model to determine the potentially significant predictors. From this, we see that MD_FAMINC, ADM_RATE_SUPP, SATVRMID, SATMTMID, COSTT4_A are all potentially significant. Neither UGDS nor STUFACR were significant. Next, we ran a VIF in order to determine if any multicollinearity is present. The only major concern was between SATVRMID and SATMTMID, where their correlation was 0.932 and both had VIFs above 8.

In order to determine the best predictors to use for our model, we ran a stepwise regression. To determine the best model, we considered parsimony, Mallows’ Cp, the adjusted R^2 and the overall F-statistic. Our chosen model had a relatively low Mallows’ CP, few predictors, and a relatively high adjusted R^2 -squared of 0.521210. However, this model included SATVRMID, SATMTMID, COSTT4_A. Since we saw previously that SATVRMID and SATMTMID were multicollinear predictors ($VIF > 5$), we chose to keep only SATMTMID (Median Math SAT), as this was a stronger predictor of future earnings than the Median Verbal SAT score. Our current model includes only three predictors: COSTT4_A, SATMTMID and MD_FAMINC. With an adjusted R^2 -squared value of 0.513, our model accounts of 51.35% of variability in Median Earnings 10 Years After Enrollment (F-statistic: 449 on 2 and 876 DF, p-value: $< 2e-16$). Additionally, we ran a nested F-test which confirmed that our three-predictor model was superior to models with more predictors.

```
##
## Call:
```

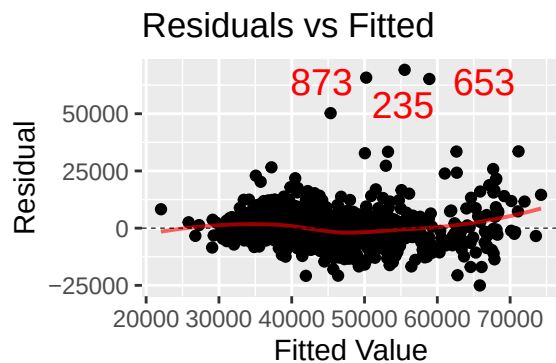
```
## lm(formula = MD_EARN_WNE_P10 ~ ., data = college_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23880  -4577   -943    3202   67887
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.921e+03  3.774e+03  -0.509  0.61081
## MD_FAMINC      8.993e-02  1.868e-02   4.813 1.75e-06 ***
## STUFACR       1.426e+02  9.551e+01   1.493  0.13592
## ADM_RATE_SUPP -4.339e+03  1.680e+03  -2.583  0.00995 **
## SATVRMID     -6.755e+01  1.158e+01  -5.833 7.65e-09 ***
## SATMTMID      1.353e+02  1.108e+01  12.211 < 2e-16 ***
## COSTT4_A      1.987e-01  3.130e-02   6.349 3.47e-10 ***
## UGDS          7.522e-02  4.940e-02   1.523  0.12820
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8234 on 871 degrees of freedom
## Multiple R-squared:  0.5388, Adjusted R-squared:  0.5351
## F-statistic: 145.4 on 7 and 871 DF, p-value: < 2.2e-16

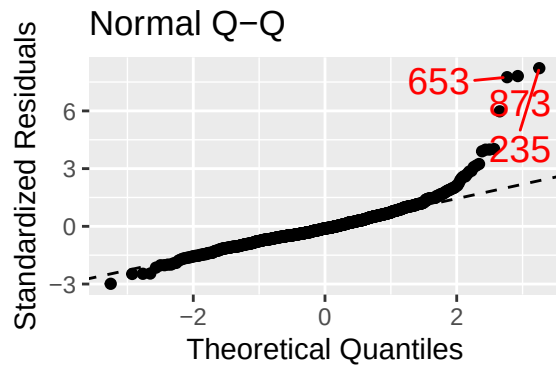
##      MD_FAMINC      STUFACR ADM_RATE_SUPP      SATVRMID      SATMTMID
##      1.852869      2.036980      1.383684      8.251630      8.650175
##      COSTT4_A      UGDS
##      2.347928      1.900669

## SATMTMID COSTT4_A
## 1.340246 1.340246
```

Conditions Checking our conditions, we see that our Residuals vs Fitted plot appears to be roughly centered about 0, with no visible heteroskedasticity. Thus, we can conclude that linearity, homoschedasticity, and independence of errors conditions are met. Looking at our QQ-plot, we see that the lower and middle parts of the data are relatively normal, however our upper tail contains several influential points that may be outliers. Further analysis involving Cook's distance will be conducted to determine if these influential points can be removed, and if they share any relation with each other. However, despite the problematic upper tail, we can reasonably assume that this condition is met.

```
## `geom_smooth()` using formula = 'y ~ x'
```





Current Conclusions Our major conclusion at this point is that a three predictor model (Median SAT score, Cost of Attendance, and Family Income) accounts for approximately 51% of the variability in median earnings 10 years after enrollment, despite the large number of potentially significant predictors hypothesized. Additionally, we found that the points with the largest deviation from the QQ line were all pharmaceutical colleges. These schools had high acceptance rates, low SAT scores, and very high median earnings, making them very unique cases. We will further investigate this phenomenon in our final submission.

Additional Analyses & Next Steps: First, we will analyze Cook's Distance and potentially remove outliers to create a better fitting model. Additionally, in our next phase, we will test our cross-validation efforts to estimate the model's true out-of-sample performance, which we anticipate to be lower than the training performance. We will calculate and report the shrinkage, and we will also test the accuracy of our predictions on several schools nationwide, including Amherst College and UMASS Amherst. Furthermore, we will use bootstrapping to try to account for the fact that cost of attendance does not have a perfectly linear relationship with median earnings 10 years after enrollment.

```
## Analysis of Variance Table
##
## Model 1: MD_EARN_WNE_P10 ~ SATMTMID
## Model 2: MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A
## Model 3: MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC
## Model 4: MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC + ADM_RATE_SUPP
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1      877 6.6155e+10
## 2      876 6.3215e+10  1 2940040576 41.4597 1.983e-10 ***
## 3      875 6.2163e+10  1 1052165481 14.8374 0.0001258 ***
## 4      874 6.1978e+10  1  184554003  2.6025 0.1070542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```